TAYYBA ABID

AI & Full-Stack Web Developer | Generative AI | RAG Systems | LangChain

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PROFESSIONAL SUMMARY

AI & Full-Stack Developer with hands-on experience architecting Generative AI applications, RAG-based document chatbots, and LLM-powered SaaS platforms. Proficient in Python, FastAPI, LangChain, and vector databases, with a strong foundation in prompt engineering and scalable backend design. Final-year Software Engineering student (CGPA 3.53) at COMSATS University, actively building production-ready AI systems. Seeking to contribute as a developer who bridges cutting-edge AI research with real-world product delivery.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, RAG Architecture, Prompt Engineering, Fine-Tuning, Hugging Face Transformers, Vector Databases (FAISS, Pinecone), Chatbot Development

Web & Backend: FastAPI, Django, Streamlit, REST APIs, Microservices Architecture, WordPress, Bootstrap, Tailwind CSS

Programming Languages: Python, JavaScript, SQL, R, C++, HTML/CSS

NLP: Text Classification, Sentiment Analysis, Document Parsing (PyMuPDF, python-docx, python-pptx)

DevOps & Tools: Docker, Git & GitHub, Jupyter Notebook, Google Colab, VS Code, Hugging Face Hub

PROJECTS

AI Learning Suite (Final Year Project) | COMSATS University *2024 – Present*

Technologies: *FastAPI, Streamlit, LangChain, RAG, Vector Databases, PyMuPDF, python-docx, python-pptx*

- Engineered a dual-application AI tutoring platform (Code Mentor AI + Quiz Generator) consuming shared FastAPI endpoints, reducing backend duplication by centralizing LLM orchestration across both products.
- Built an adaptive Code Mentor AI that auto-detects user proficiency (beginner / intermediate / advanced) via multi-turn LLM classification and dynamically adjusts response depth — achieving context-aware programming Q&A with file upload support.
- Developed auto-quiz generation pipeline that triggers after every 6 user queries, automatically producing a formatted Word document quiz + answer key, streamlining instructor assessment workflows.
- Implemented a multi-format Quiz Generator accepting PDF, DOCX, and PPT uploads; chunks and indexes content into a vector store, then generates difficulty-tiered MCQ / Short / Long answer quizzes with AI refinement — exported as PDF, DOCX, or PPT.

Multi-Bot Chat Base System | Personal / SaaS Project *2024*

Technologies: *FastAPI, LangChain, RAG, Vector Databases, Python*

- Architected a Chatbase.co-inspired SaaS platform enabling multiple users to upload proprietary documents and spin up isolated, document-scoped AI chatbots — each bot strictly grounded in its own data with zero cross-contamination.
- Implemented per-user, per-bot vector store isolation, ensuring complete data privacy and session separation across concurrent users.
- Designed RESTful FastAPI endpoints following microservices principles, enabling horizontal scaling of individual chatbot instances.

Smart PDF Chat (RAG) | Personal Project *2024*

Technologies: *LangChain, RAG, Streamlit, Vector Databases, Python*

- Built an end-to-end RAG pipeline enabling users to upload PDF documents and receive AI-generated answers strictly grounded in source content — eliminating hallucination beyond provided document scope.
- Deployed with Streamlit for an intuitive UI; integrated document chunking, embedding generation, and semantic retrieval into a seamless single-page experience.

WordPress Web Development | Freelance / Client Projects *2023 – 2024*

Technologies: *WordPress, Bootstrap, Tailwind CSS, PHP, WooCommerce*

- Delivered 3 production websites: POLOPOINT (corporate business site with responsive design), Al-Rafiq Mart (e-commerce store with WooCommerce product management), and Connect Intermodal (corporate logistics platform).
- Implemented responsive UI with Bootstrap and Tailwind CSS, ensuring cross-device compatibility and optimized page load performance.

EDUCATION

BS in Software Engineering | COMSATS University Islamabad *2022 – 2026*

CGPA: 3.53 / 4.00

Relevant Coursework: Artificial intelligence , Computer visions, Intro to Data Science, Database Systems, Software Architecture, Web Engineering

KEY ACHIEVEMENTS

- Independently designed and deployed 3+ production-grade AI applications integrating LLMs, vector search, and full-stack web frameworks.
- Maintained a 3.53 CGPA while simultaneously building a professional AI/web development portfolio with real-world client projects.
- Contributed to open-source-style AI tooling that addresses practical gaps in adaptive learning and document intelligence.